

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
Mini-Term Test (13-February-2025)

Course : B.Tech.

Semester-IV

Branch : Electrical

Subject : Engg. Mathematics IV

Subject Code: MTH241

Max.Marks : 10

Time

: 45 Minutes

Marks ↓

Note: Attempt all the questions.

1.	Find all basic solutions of the system of equations $x + y + z = 8, 2x + y + 2z = 10, x, y, z \geq 0$. Also identify those basic solutions which are basic feasible. Are there any degenerate basic feasible solution?	C01	2
2.	Use the graphical method to solve the following LP problem: <i>Minimum</i> $Z = 20x + 10y$ Subject to the constraints $x + 2y \leq 40, 3x + y \geq 30, 4x + 3y \geq 60$ and $x, y \geq 0$.	C01	2
3.	Use simplex method to solve the following LP problem: Maximize $Z = 4x_1 + x_2$ Subject to the constraints: $x_1 - x_2 \leq 1, -2x_1 + x_2 \leq 2$ and $x_1, x_2 \geq 0$.	C02	3
4.	Use the two-phase simplex method to solve the following LPP Min. $Z = x_1 - x_2$ subject to the constraints: $x_1 + x_2 \leq 1, 2x_1 + 3x_2 \geq 6, x_1, x_2 \geq 0$.	C02	3